

1. PROJECT OVERVIEW

- **Project Title:** ComicCraft - AI Comic Story Creator using Gemini Models

2. DEPLOYMENT ENVIRONMENT :

Unlike traditional desktop software, ComicCraft is engineered as a dynamic web application driven by the FastAPI framework. It does not require compilation into a standalone executable file (e.g., .exe). Instead, the platform is hosted locally within a lightweight Python Virtual Environment, making it highly portable and compatible across Windows, macOS, and Linux operating systems.

3. SYSTEM PREREQUISITES

- **Base Interpreter:** Python 3.10+ must be installed and properly added to the system PATH.
- **Operating System:** Cross-platform (Windows 10/11 is recommended for this specific test configuration).
- **Network Dependency:** A high-speed, stable internet connection is strictly required to facilitate the continuous data exchange with the cloud-based AI endpoints (Google Gemini and Hugging Face).

4. RUNTIME EXECUTION INSTRUCTIONS

Step 1: Access the Source Directory Extract the downloaded project ZIP archive. Open your preferred command-line interface (PowerShell or Terminal) and navigate to the root directory where the `main.py` file resides.

Step 2: Virtual Environment Setup To ensure that project dependencies do not conflict with your global Python installation, create and activate an isolated virtual workspace:

```
python -m venv env
.\env\Scripts\activate
```

Step 3: Dependency Installation Install the required core libraries (including fastapi, uvicorn, pillow, fpdf, and python-dotenv) utilizing the provided requirements file:

```
pip install -r requirements.txt
```

Step 4: Configure Environment Variables Create a hidden file named `.env` in the root folder. Inject your active cloud API keys here to authorize external network requests:

```
GEMINI_API_KEY="your_google_studio_key"  
HF_API_KEY="your_hugging_face_token"
```

Step 5: Start the ASGI Web Server Boot up the backend application logic using the Uvicorn server:

```
uvicorn app.main:app --reload
```

Step 6: Access the Frontend Interface Open a modern web browser and enter the following local host address to begin interacting with the ComicCraft platform:

<http://127.0.0.1:8000>

5. RUNTIME TROUBLESHOOTING

- **Activation Failure (UnauthorizedAccess):** If Windows PowerShell blocks the activation of the virtual environment, the system's execution policy must be bypassed by running `Set-ExecutionPolicy Unrestricted -Scope CurrentUser`.
- **Path Length Limitation (WinError 206):** This occurs during `pip install` if the operating system restricts lengthy file paths. It can be

resolved by enabling `LongPathsEnabled` within the Windows Registry Editor.

- **Missing Dependencies (`ModuleNotFoundError`):** If the server crashes immediately upon launch, the virtual environment did not capture the installed libraries correctly. Delete the `env` folder, recreate it, activate it securely, and rerun the installation command.
- **Network Timeout (`getaddrinfo failed`):** If local internet service providers block API access to Hugging Face, the application will not crash. The backend is designed to intercept this failure automatically and generate a localized PIL placeholder image, allowing the PDF compilation phase to finish seamlessly.